ROADSURE KNOWLEDGE BASE

RoadSure – Road Quality Insurance Platform

Natural-language knowledge source for retrieval-augmented generation (RAG)

Version 1.0 · Prepared 19 October 2026

Includes database-backed records, clearly labelled illustrative sample data, workflows, notifications, data-quality notes, and 30 FAQs.

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Version: 1.0 Prepared: 19 October 2026 Purpose: Natural-language

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HOW TO USE THIS KNOWLEDGE BASE

This file describes the RoadSure road-quality insurance platform. It is written as natural-language knowledge rather than as a raw database dump so that a RAG system can retrieve coherent facts, workflows, and explanations.

DATA LABELS AND ACCURACY - DATABASE RECORD: Information transcribed from

the database output supplied for this knowledge base. Recorded values

and dates are preserved. - ILLUSTRATIVE SAMPLE: Realistic example data

added to make the knowledge base more useful. It is not a claim that the

record exists in the live database. - BUSINESS RULE / PROPOSED WORKFLOW:

A sensible description of how RoadSure may operate. Unless specifically supported by the supplied records, it should not be represented as a

verified production rule. - DATA QUALITY NOTE: A value that appears

incomplete, unusual, contradictory, or potentially erroneous. Preserve it for traceability, but do not treat it as a reliable business conclusion without verification.

RoadSure provides insurance for eligible premature road failures attributable to covered construction, material, design, or quality risks. It does not mean that every road defect or every repair cost is automatically insured. Normal wear and tear, routine maintenance, and damage outside the agreed coverage should not be assumed to be covered. Actual coverage depends on the policy wording, exclusions, evidence, and claim assessment.

Security note: Password hashes from the users table are deliberately

omitted. They are not needed for retrieval and should never be included in a user-facing knowledge document.

========== 1.

ORGANIZATIONS

DATABASE RECORD: ORGANIZATION 1 Organization ID: 1 Organization name:

Government Road Authority Organization type: ROAD_AUTHORITY Address:

Hyderabad Contact email: gov@test.com Contact number: 9876543210 Status:

ACTIVE Created at: 2026-09-16 12:15:48 Updated at: 2026-09-16 12:15:48

Description: Government Road Authority represents the government-side

road authority. It owns or oversees road projects, establishes project requirements, reviews construction progress, and monitors contractor performance. In the RoadSure workflow, the road authority can create or manage road projects, review inspection outcomes, and raise insurance claims as the policyholder when an eligible covered loss occurs.

DATABASE RECORD: ORGANIZATION 2 Organization ID: 2 Organization name:

ABC Infrastructure Pvt Ltd Organization type: CONTRACTOR Address:

Hyderabad Contact email: contractor@test.com Contact number: 9876543211

Status: ACTIVE Created at: 2026-09-16 12:16:07 Updated at: 2026-09-16

12:16:07

Description: ABC Infrastructure Pvt Ltd is a road construction

contractor. It is associated with the NH 47 Road Expansion project. The contractor submits construction details, records progress, responds to inspection findings, and carries out corrective work when a defect is identified.

DATABASE RECORD: ORGANIZATION 3 Organization ID: 3 Organization name:

Hartford Insurance Organization type: INSURANCE_COMPANY Address:

Hyderabad Contact email: insurance@test.com Contact number: 9876543212

Status: ACTIVE Created at: 2026-09-16 12:16:17 Updated at: 2026-09-16

12:16:17

Description: Hartford Insurance is the insurance organization in the

supplied RoadSure sample data. It administers the platform, assesses project risks, prepares policy proposals, issues policies, reviews claims, and processes approved settlements. The platform's System Admin, Insurance Underwriter, and Claims Adjuster users belong to this organization.

DATABASE RECORD: ORGANIZATION 4 Organization ID: 4 Organization name:

Road Quality Labs Organization type: INSPECTION_ORGANIZATION Address:

Hyderabad Contact email: inspection@test.com Contact number: 9876543213

Status: ACTIVE Created at: 2026-09-16 12:16:24 Updated at: 2026-09-16

12:16:24

Description: Road Quality Labs is the inspection organization. Its

inspector records initial, periodic, and final inspections, including material-quality scores, layer-thickness scores, test scores, remarks, and inspection outcomes. It also records defects and verifies corrective actions.

DATABASE RECORD: ORGANIZATION 5 Organization ID: 5 Organization name:

RoadCorp Contractors Organization type: CONTRACTOR Address: 2-135/19,

Hyderabad Contact email: roadcorp@test.com Contact number: 9876543333

Status: ACTIVE Created at: 2026-09-21 13:12:55 Updated at: 2026-09-21

13:12:55

Description: RoadCorp Contractors is a second contractor organization in

the RoadSure sample environment. It is associated with the illustrative Outer Ring Road Smart Corridor project in this knowledge base. It is responsible for construction execution, progress updates, and addressing inspection findings for projects assigned to it.

2. USERS AND ROLES

DATABASE RECORD: USER 1 — SYSTEM ADMIN User ID: 1 Name: System Admin

Email: admin@test.com Role: ADMIN Organization: Hartford Insurance

Organization ID: 3 Status: ACTIVE Created at: 2026-09-16 12:17:48

Updated at: 2026-10-07 16:41:08

Responsibilities: The System Admin manages platform administration, user

accounts, organization records, and general system oversight. Administrative access should be used for platform management rather than for making technical inspection judgments or independently determining insurance losses.

DATABASE RECORD: USER 2 — ROAD AUTHORITY PM User ID: 2 Name: Road

Authority Email: roadauthority@test.com Role: ROAD_AUTHORITY_PM

Organization: Government Road Authority Organization ID: 1 Status:

ACTIVE Created at: 2026-09-16 12:18:10 Updated at: 2026-09-16 12:18:10

Responsibilities: The Road Authority Project Manager represents the

government-side authority. This user creates or manages projects, reviews progress and inspection outcomes, and raises claims against the relevant policy when an eligible covered road failure occurs. In the supplied records, this user created the NH 47 project and raised both recorded claims.

DATABASE RECORD: USER 4 — CONTRACTOR PM User ID: 4 Name: Contractor PM

Email: contractorpm@test.com Role: CONTRACTOR_PM Organization: ABC

Infrastructure Pvt Ltd Organization ID: 2 Status: ACTIVE Created at:

2026-09-16 12:20:23 Updated at: 2026-09-23 09:33:29

Responsibilities: The Contractor Project Manager submits construction

details, updates project progress, coordinates construction execution, and responds to defects and inspection findings. This user created the recorded progress updates for the NH 47 project.

DATABASE RECORD: USER 5 — INSURANCE UNDERWRITER User ID: 5 Name:

Insurance Underwriter Email: underwriter@test.com Role: UNDERWRITER

Organization: Hartford Insurance Organization ID: 3 Status: ACTIVE

Created at: 2026-09-16 12:21:04 Updated at: 2026-09-16 12:21:04

Responsibilities: The Insurance Underwriter assesses the risk associated

with a road project, considers construction quality and other risk factors, and generates policy proposals. The supplied risk assessment and policy proposal were recorded against this user.

DATABASE RECORD: USER 6 — CLAIMS ADJUSTER User ID: 6 Name: Claims

Adjuster Email: adjuster@test.com Role: CLAIMS_ADJUSTER Organization:

Hartford Insurance Organization ID: 3 Status: ACTIVE Created at:

2026-09-16 12:21:19 Updated at: 2026-09-16 12:21:19

Responsibilities: The Claims Adjuster reviews submitted claims, examines

evidence, assesses severity and eligible loss, recommends a settlement, and records approval or settlement actions. In the supplied records, this user completed both claim assessments and approved the corresponding settlements.

DATABASE RECORD: USER 7 — ROAD INSPECTOR User ID: 7 Name: Road Inspector

Email: inspector@test.com Role: INSPECTOR Organization: Road Quality

Labs Organization ID: 4 Status: ACTIVE Created at: 2026-09-16 12:21:48

Updated at: 2026-09-16 12:21:48

Responsibilities: The Road Inspector performs initial, periodic, and

final inspections. The inspector records test scores and remarks, identifies defects, and verifies whether corrective actions have resolved the reported issue.

DATABASE RECORD: USER 8 — ROADCORP CONTRACTOR PM User ID: 8 Name: Road

Corp Email: roadcorp@test.com Role: CONTRACTOR_PM Organization: RoadCorp

Contractors Organization ID: 5 Status: ACTIVE Created at: 2026-09-21

13:14:18 Updated at: 2026-09-23 10:21:10

Responsibilities: This Contractor Project Manager represents RoadCorp

Contractors and is responsible for execution and progress reporting on projects assigned to that organization. In the illustrative data, this user is associated with the Outer Ring Road Smart Corridor project.

Role separation: - ADMIN: Platform and account administration. -

ROAD_AUTHORITY_PM: Government-side project management and claim

submission. - CONTRACTOR_PM: Construction details, progress reporting,

and corrective work. - UNDERWRITER: Project risk assessment and policy

proposal preparation. - CLAIMS_ADJUSTER: Claim assessment and settlement

processing. - INSPECTOR: Technical inspections, defect reporting, and

corrective-action verification.

Access should be restricted by role and organization. A contractor should not be assumed to have authority to approve its own inspection results, issue insurance policies, or approve claim settlements.

3. ROAD PROJECTS

DATABASE RECORD: PROJECT 1 — NH 47 ROAD EXPANSION Project ID: 1 Project

name: NH 47 Road Expansion Description in database: The NH 47 is in its

maximum capacity, It is being expanded to 8 Lanes. Location: Chennai

Road length: 100 km Estimated cost: INR 1,000,000,000 Expected lifespan:

50 years Road authority organization ID: 1 — Government Road Authority

Contractor organization ID: 2 — ABC Infrastructure Pvt Ltd Start date:

2026-10-09 Estimated completion date recorded: 2029-07-25 Project status

recorded: COMPLETED Created by user ID: 2 — Road Authority PM Created

at: 2026-10-09 09:53:29 Updated at: 2026-10-09 10:17:22

Natural-language description: The NH 47 Road Expansion project was

initiated because the existing highway was operating at its capacity. The planned objective was to expand the road to an eight-lane corridor to accommodate increasing traffic demand. The project record lists a length of 100 kilometres, an estimated cost of INR 1 billion, and an expected lifespan of 50 years. ABC Infrastructure Pvt Ltd is the contractor, and Government Road Authority is the road authority.

The construction details record an asphalt-based, phased approach. Site preparation and surveying are followed by subgrade stabilization, aggregate base installation, asphalt paving, compaction, and final quality inspections. The progress records show updates at 20%, 40%, 75%, and 100%. The final project status is recorded as COMPLETED.

Data quality note: The project is marked COMPLETED and progress reached

100% in records dated 2026-10-09, while the estimated completion date is 2029-07-25. Preserve both database values. Do not infer why they differ. The supplied inspection dates also include dates in 2027, after the database creation timestamps; those dates are preserved as recorded.

ILLUSTRATIVE SAMPLE: PROJECT 2 — OUTER RING ROAD SMART CORRIDOR Project

ID: 2 Project name: Outer Ring Road Smart Corridor Location: Hyderabad

Road length: 75 km Estimated cost: INR 850,000,000 Expected lifespan: 40

years Road authority: Government Road Authority Contractor: RoadCorp

Contractors Proposed status: UNDER_CONSTRUCTION Illustrative start date:

2026-08-01 Illustrative expected completion date: 2028-12-31

Description: The Outer Ring Road Smart Corridor is an illustrative road

development project in Hyderabad. It aims to improve traffic flow through a modern road corridor that includes smart traffic systems, improved drainage infrastructure, and modern road-safety mechanisms. RoadCorp Contractors is responsible for construction execution, while Government Road Authority oversees project requirements and progress.

Illustrative construction approach: Work begins with a corridor survey,

utility mapping, traffic-diversion planning, and drainage assessment. The contractor then prepares the subgrade, installs base layers, constructs pavement sections, installs road-safety features, and integrates smart traffic equipment. Inspections are scheduled at material approval, base-layer completion, pavement completion, and final handover.

Illustrative progress: As of the example record, project progress is

35%. Utility mapping and initial drainage work are complete, and selected road sections are undergoing base preparation. These details are sample data, not records extracted from the provided database.

Illustrative risk considerations: Risk reviewers should consider

drainage performance, pavement material consistency, traffic diversion, workmanship, equipment integration, and compliance with approved specifications. A high drainage score or low material-quality score should trigger further review rather than an automatic coverage decision.

ILLUSTRATIVE SAMPLE: PROJECT 3 — COASTAL HIGHWAY REHABILITATION Project

ID: 3 Project name: Coastal Highway Rehabilitation Location:

Visakhapatnam Road length: 55 km Estimated cost: INR 500,000,000

Expected lifespan: 35 years Road authority: Government Road Authority

Contractor: ABC Infrastructure Pvt Ltd Proposed status:

INSPECTION_PENDING Illustrative start date: 2026-07-15 Illustrative

expected completion date: 2027-12-31

Description: The Coastal Highway Rehabilitation project focuses on

rehabilitating damaged coastal roads affected by weather exposure and heavy transport. The illustrative scope includes evaluating pavement distress, improving drainage, repairing eligible structural defects, replacing unsuitable materials where justified, and restoring road-safety features. ABC Infrastructure Pvt Ltd executes the work, and Government Road Authority oversees the project.

Illustrative construction approach: The contractor first documents

existing road conditions and identifies sections requiring rehabilitation. Damaged pavement is removed where necessary, the base and subgrade are assessed, drainage deficiencies are addressed, and suitable materials are installed and compacted. A technical inspection is required before the rehabilitated section is accepted.

Illustrative status: The project is marked INSPECTION_PENDING in this

example because rehabilitation work has been submitted for a quality review but the required inspection approval has not yet been recorded. This is a proposed sample scenario, not a fact from the supplied live database.

Illustrative risk considerations: Coastal exposure, water ingress,

drainage adequacy, heavy vehicle loading, material selection, and construction workmanship may influence project risk. Weather-related damage is not automatically insured. A claim must be assessed against the policy’s covered causes, exclusions, evidence, and terms.

4. CONSTRUCTION DETAILS

DATABASE RECORD: NH 47 CONSTRUCTION DETAILS Construction detail ID: 1

Project ID: 1 — NH 47 Road Expansion Material type: ASPHALT Created at:

2026-10-09 09:55:22 Updated at: 2026-10-09 09:55:22 Timeline recorded: 2

Years Equipment details: Pavers, Rollers, Movers, Pipes Workforce

details: 200 People

Methodology: The project uses a phased construction approach. Site

preparation and surveying are completed first, followed by subgrade stabilization, aggregate base-layer installation, asphalt paving, compaction, and final quality inspections. Work is carried out in sections to minimize traffic disruption and maintain safety compliance.

Interpretation: Asphalt is the recorded primary material type. Pavers

support asphalt placement, rollers support compaction, and pipes are consistent with drainage or utility work. The database lists “Movers” as equipment; the term is preserved as entered rather than silently changed to a different machine type. The recorded workforce is 200 people, and the recorded construction timeline is two years.

ILLUSTRATIVE CONSTRUCTION DETAILS FOR PROJECT 2 The Outer Ring Road Smart Corridor uses staged earthwork, subgrade preparation, aggregate base installation, pavement construction, drainage installation, road markings, safety barriers, and smart traffic equipment integration. The illustrative equipment includes excavators, graders, pavers, rollers, dump trucks, drainage-laying equipment, and traffic-system installation tools. An illustrative workforce consists of civil engineers, site supervisors, machine operators, surveyors, safety personnel, skilled workers, and general labourers.

ILLUSTRATIVE CONSTRUCTION DETAILS FOR PROJECT 3 The Coastal Highway Rehabilitation project includes pavement removal where required, subgrade and base assessment, drainage improvements, approved material replacement, asphalt resurfacing, compaction, and safety-feature restoration. Sample equipment includes milling machines, excavators, graders, pavers, rollers, water pumps, and compaction-testing equipment. The contractor should maintain evidence of material batches, test results, weather conditions, and work completed in each road section.

5. CONSTRUCTION PROGRESS

DATABASE RECORD: NH 47 PROGRESS HISTORY

Progress record 1: On 2026-10-09, project 1 was reported at 20%

completion. The work description states that excavation was completed. The record was created by user ID 4, the Contractor PM, at 2026-10-09

09:57:29.

Progress record 2: On 2026-10-09, project 1 was reported at 40%

completion. The work description states that the pipeline had been laid

out. The record was created by user ID 4 at 2026-10-09 09:57:53.

Progress record 3: On 2026-10-09, project 1 was reported at 75%

completion. The work description states that road laying had started.

The record was created by user ID 4 at 2026-10-09 10:09:33.

Progress record 4: On 2026-10-09, project 1 was reported at 100%

completion. The work description states that the road had been

completed. The record was created by user ID 4 at 2026-10-09 10:12:04.

Progress interpretation: The supplied progress history moves from 20% to

40%, then 75%, and finally 100%. The entries show excavation, pipeline layout, road laying, and road completion. Progress percentage is a reported construction milestone; it does not by itself prove that all inspections, contractual acceptance, or insurance obligations are complete. Those states should be checked separately.

ILLUSTRATIVE PROGRESS FOR PROJECT 2 The Outer Ring Road Smart Corridor is at 35% in the sample scenario. Surveying, utility mapping, and some drainage activities have been completed. Base preparation is ongoing in selected sections. The next milestones are completion of drainage work, base-layer inspection, pavement placement, and smart traffic system integration.

ILLUSTRATIVE PROGRESS FOR PROJECT 3 The Coastal Highway Rehabilitation sample is at 80% construction completion, but a quality inspection is pending. The contractor has completed major rehabilitation work and submitted evidence for review. The road authority should not treat the project as fully accepted until the applicable inspection and approval process is complete.

6. INSPECTIONS

DATABASE RECORD: INSPECTION 1 — INITIAL INSPECTION Inspection ID: 1

Project: NH 47 Road Expansion Inspection type: INITIAL Inspector user

ID: 7 — Road Inspector Inspection date recorded: 2026-10-16

Material-quality score: 90 Layer-thickness score: 85 Test score: 92

Inspection status: PASSED Approval status: APPROVED Created at:

2026-10-09 09:56:06 Remarks: “The site is good and all tests have been

passed.”

Narrative: The initial inspection of NH 47 was recorded as passed and

approved. The material-quality score was 90, the layer-thickness score was 85, and the test score was 92. The inspector’s remarks state that the site was good and all tests had passed. This record indicates an approved initial quality check in the supplied database.

DATABASE RECORD: INSPECTION 2 — PERIODIC INSPECTION Inspection ID: 2

Project: NH 47 Road Expansion Inspection type: PERIODIC Inspector user

ID: 7 — Road Inspector Inspection date recorded: 2027-01-15

Material-quality score: 40 Layer-thickness score: 90 Test score: 50

Inspection status: FAILED Approval status: REJECTED Created at:

2026-10-09 09:58:51 Remarks: “The new material was not on standards of

the tests.”

Narrative: The first recorded periodic inspection failed and was

rejected. The material-quality score was 40, the layer-thickness score was 90, and the test score was 50. The inspector reported that the new material did not meet the test standards. A material defect was then recorded against inspection 2. The failure indicates that a section required investigation and corrective action; it should not be interpreted as proof that every part of the project was defective.

DATABASE RECORD: INSPECTION 3 — PERIODIC INSPECTION AFTER CORRECTIVE

WORK Inspection ID: 3 Project: NH 47 Road Expansion Inspection type:

PERIODIC Inspector user ID: 7 — Road Inspector Inspection date recorded:

2027-01-13 Material-quality score: 90 Layer-thickness score: 90 Test

score: 90 Inspection status: PASSED Approval status: APPROVED Created

at: 2026-10-09 10:06:02 Remarks: “All the test have been passed with

resolution for the defect.”

Narrative: The third inspection record is a periodic inspection marked

passed and approved. Material quality, layer thickness, and test scores are all 90. The remarks state that all tests passed and the defect had been resolved. The record is consistent with the separate corrective-action entry, which states that the material was changed and verification status was VERIFIED. The inspection date is recorded as 2027-01-13, which is earlier than the 2027-01-15 date on inspection 2. Both dates are preserved as supplied; the ordering should be checked in the source system if chronology matters.

DATABASE RECORD: INSPECTION 4 — PERIODIC INSPECTION Inspection ID: 4

Project: NH 47 Road Expansion Inspection type: PERIODIC Inspector user

ID: 7 — Road Inspector Inspection date recorded: 2027-03-09

Material-quality score: 90 Layer-thickness score: 80 Test score: 90

Inspection status: PASSED Approval status: APPROVED Created at:

2026-10-09 10:11:19 Remarks: “None.”

Narrative: A later periodic inspection was recorded as passed and

approved. The material-quality score was 90, the layer-thickness score was 80, and the test score was 90. No substantive remarks were provided in the database. The inspection therefore records a successful outcome, while the layer-thickness score remains a separate technical measure that may be interpreted against the project’s actual acceptance criteria.

DATABASE RECORD: INSPECTION 5 — FINAL INSPECTION Inspection ID: 5

Project: NH 47 Road Expansion Inspection type: FINAL Inspector user ID:

7 — Road Inspector Inspection date recorded: 2027-08-09 Material-quality

score: 90 Layer-thickness score: 90 Test score: 92 Inspection status:

PASSED Approval status: APPROVED Created at: 2026-10-09 10:12:44

Remarks: “All the tests have been passed with standard results.”

Narrative: The final inspection was recorded as passed and approved. The

material-quality score and layer-thickness score were both 90, and the test score was 92. The inspector stated that all tests passed with standard results. This is the final inspection record supplied for NH 47 and is consistent with a successful final quality check in the database.

INSPECTION GUIDANCE An initial inspection establishes whether the work or materials meet the initial acceptance requirements. Periodic inspections monitor quality as construction proceeds and can identify defects that need correction. A

final inspection evaluates the completed work against the relevant acceptance criteria. A PASSED inspection status and an APPROVED approval status are related but distinct fields in the database.

A FAILED or REJECTED inspection should lead to review of the findings, documentation of any defect, corrective action by the responsible contractor, and verification by an authorized inspector. A corrected defect should not be considered verified merely because the contractor says the work is done; the verification result should be recorded.

No universal pass-score threshold is specified in the supplied records. Do not invent a rule such as “every score must be above 80” unless the application’s configured business rules confirm it.

ILLUSTRATIVE INSPECTION SCENARIOS FOR PROJECTS 2 AND 3 For the Outer Ring Road Smart Corridor, an illustrative base-layer inspection records material quality of 88, layer thickness of 86, and test score of 91, with a passed outcome. These numbers are sample data and are not extracted from the database.

For the Coastal Highway Rehabilitation project, an illustrative pending inspection requests additional evidence about drainage compaction and asphalt quality before acceptance. The sample status remains INSPECTION_PENDING until the inspection outcome is recorded. These details illustrate how a pending inspection might be represented; they are not actual database records.

7. DEFECTS

DATABASE RECORD: DEFECT 1 Defect ID: 1 Project ID: 1 — NH 47 Road

Expansion Inspection ID: 2 Defect type: Material Defect Description:

“Material quality is bad.” Status: RESOLVED Created at: 2026-10-09

09:59:26

Narrative: A material defect was recorded for NH 47 following inspection

2. The description says the material quality was bad, and the defect is

linked to the periodic inspection that failed and was rejected. The defect status is RESOLVED. The corrective-action record states that the material was changed, and its verification status is VERIFIED.

Defect management guidance: A defect record should identify the affected

project and inspection, describe the issue clearly, and record the current status. Useful defect evidence may include photographs, test results, material batch identifiers, location coordinates or chainage, the date discovered, severity, and the responsible party. Those additional fields are recommended examples and are not all present in the supplied database.

A defect is not automatically an insurance claim. The defect may be corrected during construction, may be excluded by a policy, or may not have caused an insured loss. Insurance eligibility requires separate policy and claim review.

ILLUSTRATIVE DEFECT: PROJECT 2 A sample defect is identified during a

base-layer inspection: drainage compaction is below the approved

specification in a limited section. The contractor is asked to rework the affected area and submit new test results. This is a sample scenario, not a live database record.

ILLUSTRATIVE DEFECT: PROJECT 3 A sample inspection identifies localized

asphalt segregation near a coastal road section. The inspector requests material-batch traceability and a corrective plan. The issue is not assumed to be insured; the cause, timing, policy coverage, and actual loss would need to be established.

8. CORRECTIVE ACTIONS

DATABASE RECORD: CORRECTIVE ACTION 1 Corrective action ID: 1 Defect ID:

1 Action taken: “Material Has been changed.” Verification status:

VERIFIED Created at: 2026-10-09 10:03:27 Updated at: 2026-10-09 10:04:02

Narrative: The corrective action for the NH 47 material defect states

that the material was changed. The action is linked to defect 1, and the verification status is VERIFIED. This record supports the later inspection remarks that the defect had been resolved. The database does not specify the exact replacement material batch or attach test certificates in the output supplied.

Proposed corrective-action workflow: First, the inspector records the

defect and the evidence supporting it. Second, the contractor reviews the finding and submits a corrective plan. Third, the contractor performs the corrective work and records what was changed. Fourth, the inspector verifies the work and records whether it passes. If verification fails, the defect should remain unresolved or be reopened according to the system's configured rules. Only verified evidence should be treated as confirmation that the reported corrective work passed.

The workflow above is a proposed operational description. The supplied database confirms one corrective action with a VERIFIED status but does not show every possible state transition.

9. RISK ASSESSMENTS

DATABASE RECORD: RISK ASSESSMENT 1 Assessment ID: 1 Project: NH 47 Road

Expansion Project ID: 1 Material-quality score: 90 Layer-thickness

score: 90 Test score: 92 Defect score: 10 Traffic-load score: 50

Drainage score: 24 Risk score: 18.67 Risk category: LOW Coverage

percentage: 90% Recommended premium: INR 10,000,000 Assessed by user ID:

5 — Insurance Underwriter Created at: 2026-10-09 10:18:33

Narrative: The NH 47 risk assessment records a risk score of 18.67 and

classifies the project as LOW risk. It records material quality of 90, layer thickness of 90, and a test score of 92. The defect score is 10, traffic-load score is 50, and drainage score is 24. The recommended coverage is 90%, and the recommended premium is INR 10 million. The assessment was recorded by the Insurance Underwriter.

Interpretation: The recorded scores provide inputs for evaluating

project risk. They should not be interpreted as percentages unless the application's scoring definition says they are percentages. The database output does not include the complete risk formula, score weighting, or thresholds used to assign risk categories. Therefore, a RAG answer may report the stored risk score and category but should not claim to reproduce the calculation without the official formula.

Proposed risk factors: - Material quality and material-batch

traceability. - Pavement and base-layer thickness. - Laboratory and field-test results. - Number, type, severity, and resolution status of defects. - Traffic loading and expected vehicle mix. - Drainage performance and water-ingress exposure. - Construction method, supervision, and compliance with specifications. - Inspection history and evidence of corrective work. - Environmental and site conditions where relevant.

A higher risk score may indicate greater risk only if that is how the application defines the score. Coverage and premium decisions should be based on the configured underwriting rules and the actual policy terms.

ILLUSTRATIVE RISK ASSESSMENT: PROJECT 2 The Outer Ring Road Smart

Corridor is assigned an illustrative MODERATE risk category because construction is ongoing and drainage, pavement, and smart traffic components still require verification. An illustrative risk score of 32.5, coverage of 85%, and recommended premium of INR 8,500,000 are sample values for demonstration only. They are not calculated from a confirmed RoadSure formula and must not be presented as actual underwriting decisions.

ILLUSTRATIVE RISK ASSESSMENT: PROJECT 3 The Coastal Highway

Rehabilitation project is assigned an illustrative MODERATE risk category because coastal exposure, drainage conditions, and heavy transport require close assessment. An illustrative risk score of 38.0, coverage of 80%, and recommended premium of INR 5,500,000 are sample values only. A real underwriting decision would require project evidence, the official scoring model, and the proposed policy terms.

10. POLICY PROPOSALS

DATABASE RECORD: POLICY PROPOSAL 1 Proposal ID: 1 Project: NH 47 Road

Expansion Assessment ID: 1 Premium amount: INR 10,000,000 Coverage

percentage: 90% Proposal status: ACCEPTED Valid until: 2028-01-09

Generated by user ID: 5 — Insurance Underwriter Created at: 2026-10-09

10:18:56 Rejection reason: NULL

Narrative: The Insurance Underwriter generated policy proposal 1 for NH

47 based on risk assessment 1. The proposal records a premium of INR 10 million and coverage of 90%. Its status is ACCEPTED, and its recorded validity date is 2028-01-09. No rejection reason is recorded because the proposal was accepted.

A proposal and an issued policy are separate records. An accepted proposal indicates that the proposal was accepted in the supplied data, while the issued policy record contains the policy number, issue date, expiry date, and active status.

11. INSURANCE POLICIES

DATABASE RECORD: POLICY 1 Policy ID: 1 Project: NH 47 Road Expansion

Proposal ID: 1 Policy number: POL-11E7B1AB Premium amount: INR

10,000,000 Coverage percentage: 90% Policy status: ACTIVE Issue date:

2026-10-09 Expiry date: 2027-10-09 Issued by user ID: 2 in the supplied

database Created at: 2026-10-09 10:19:12

Important data-quality note: The issued_by field contains user ID 2,

which maps to the Road Authority PM, rather than the Insurance Underwriter. Preserve this recorded value. Do not silently change it or assume the field's intended meaning. If the application requires policies to be issued by an authorized insurance role, verify the field mapping and permissions in the source system.

Narrative: Policy POL-11E7B1AB is recorded as ACTIVE for the NH 47 Road

Expansion project. The policy has a 90% coverage percentage, a premium amount of INR 10 million, an issue date of 2026-10-09, and an expiry date of 2027-10-09. The premium payment record for INR 10 million is marked SUCCESS on 2026-10-09.

Coverage explanation: The stored 90% coverage value should not be

interpreted as a guarantee that 90% of every repair bill will be paid. It is a recorded policy field. The actual amount payable depends on policy wording, covered causes, exclusions, deductibles or limits if applicable, evidence of loss, and the claim decision. The supplied records do not specify all those contractual terms.

RoadSure is intended to cover eligible premature road failures attributable to covered construction, material, design, or quality risks. Normal wear and tear and routine maintenance are not automatically covered. The cause of damage must be evaluated before determining coverage.

ILLUSTRATIVE POLICY PROPOSAL FOR PROJECT 2 A sample proposal for the Outer Ring Road Smart Corridor uses an illustrative premium of INR 8,500,000 and 85% coverage. The proposal remains illustrative and does not represent an actual issued policy. The policy should not be described as active unless an actual policy record confirms issuance and activation.

ILLUSTRATIVE POLICY PROPOSAL FOR PROJECT 3 A sample proposal for the Coastal Highway Rehabilitation project uses an illustrative premium of INR 5,500,000 and 80% coverage. The proposal is a demonstration scenario only. Actual coverage requires underwriting review, the official scoring model, and policy terms.

12. CLAIMS

DATABASE RECORD: CLAIM 1 Claim ID: 1 Project: NH 47 Road Expansion

Policy ID: 1 — Policy POL-11E7B1AB Claim reason: "Potholes have been

created due to heavy traffic." Estimated repair cost: INR 10,000,000

Claim status: SETTLED Raised by user ID: 2 — Road Authority PM Created

at: 2026-10-09 10:21:25 Rejection reason: NULL

Narrative: The Road Authority PM submitted claim 1 under policy

POL-11E7B1AB for potholes attributed in the claim description to heavy traffic. The estimated repair cost is INR 10 million. The claim is marked SETTLED. The associated claim assessment records MINOR severity and recommends a settlement of

INR 2.7 million. The settlement record shows INR 2.7 million with payment status PAID, and the payment table records a successful settlement payment for the same amount.

Coverage caution: The recorded claim reason mentions heavy traffic, but

this alone does not establish that the loss is covered. A real adjuster should determine whether the potholes resulted from a covered premature failure attributable to an insured construction, material, design, or quality risk, rather than normal wear, traffic exposure alone, routine maintenance, or an excluded cause. The supplied sample claim is recorded as settled, but its record does not contain the full evidence or policy wording needed to independently verify the coverage decision.

DATABASE RECORD: CLAIM 2 Claim ID: 2 Project: NH 47 Road Expansion

Policy ID: 1 — Policy POL-11E7B1AB Claim reason as entered:

“12121212112” Estimated repair cost: INR 123,456 Claim status: SETTLED

Raised by user ID: 2 — Road Authority PM Created at: 2026-10-09 12:21:53

Rejection reason: NULL

Narrative and data-quality note: Claim 2 is present in the database and

is marked SETTLED. Its reason is the numeric-looking string “12121212112,” which does not describe a meaningful insurance event. The record is preserved for traceability but should be treated as a likely test or placeholder entry. Do not use it as an example of a legitimate covered cause or infer what damage actually occurred.

13. CLAIM ASSESSMENTS

DATABASE RECORD: CLAIM ASSESSMENT 1 Assessment ID: 1 Claim ID: 1

Adjuster user ID: 6 — Claims Adjuster Severity level: MINOR Assessment

notes: “Checked the site and it is minor only.” Recommended settlement:

INR 2,700,000 Assessment status: COMPLETED Created at: 2026-10-09

10:21:51

Narrative: The Claims Adjuster completed the assessment of claim 1 and

classified the severity as MINOR. The notes say that the site was checked and the issue was minor. The recommended settlement was INR 2.7 million. The subsequent settlement record shows the same amount as PAID.

DATABASE RECORD: CLAIM ASSESSMENT 2 Assessment ID: 2 Claim ID: 2

Adjuster user ID: 6 — Claims Adjuster Severity level: MODERATE

Assessment notes as entered: “fsgdg” Recommended settlement: INR

66,666.24 Assessment status: COMPLETED Created at: 2026-10-09 12:25:22

Narrative and data-quality note: Claim assessment 2 is marked COMPLETED

with MODERATE severity and a recommended settlement of INR 66,666.24. The assessment note “fsgdg” appears to be placeholder or test text and does not explain the assessment. Preserve the recorded value, but do not use it as meaningful evidence or invent a damage explanation.

Proposed claim-assessment workflow: The adjuster verifies that the

policy was active for the relevant event, collects the claim description and supporting evidence, checks the cause of damage against the coverage terms, evaluates the severity and eligible loss, documents the reasoning, and records a recommended settlement. The claim then follows the configured approval and payment process. A settlement recommendation is not the same thing as proof of payment; the settlement and payment records should be checked separately.

14. SETTLEMENTS

DATABASE RECORD: SETTLEMENT 1 Settlement ID: 1 Claim ID: 1 Settlement

amount: INR 2,700,000 Payment status: PAID Approved by user ID: 6 —

Claims Adjuster Created at: 2026-10-09 10:23:20

Narrative: Settlement 1 is associated with claim 1. The settlement

amount is INR 2.7 million and the payment status is PAID. The database records user ID 6, the Claims Adjuster, as the approver. The payment table contains a matching successful settlement payment dated 2026-10-09.

DATABASE RECORD: SETTLEMENT 2 Settlement ID: 2 Claim ID: 2 Settlement

amount: INR 66,666.24 Payment status: PAID Approved by user ID: 6 —

Claims Adjuster Created at: 2026-10-09 12:25:34

Narrative: Settlement 2 is associated with claim 2. It records INR

66,666.24 as the settlement amount and PAID as the payment status. The amount matches the recommended settlement in claim assessment 2. Because claim 2 and its assessment contain placeholder-looking text, this settlement should also be treated as a database test record unless verified against the source application.

Settlement guidance: The settlement amount is the recorded amount

associated with the claim. It may differ from the estimated repair cost because the assessment can determine that only part of the reported cost is eligible under the policy. A valid explanation should refer to the actual assessment notes and policy terms. If those details are unavailable, the assistant should say so instead of inventing a calculation.

15. PAYMENTS

DATABASE RECORD: PAYMENT 1 — PREMIUM Payment ID: 1 Policy ID: 1 Amount:

INR 10,000,000 Payment type: PREMIUM Payment status: SUCCESS Payment

date: 2026-10-09 Created at: 2026-10-09 10:19:12

Narrative: The premium payment record shows a successful payment of INR

10 million for policy 1. It was recorded on 2026-10-09, at the same timestamp as the policy creation record.

DATABASE RECORD: PAYMENT 2 — CLAIM 1 SETTLEMENT Payment ID: 2 Policy ID:

1 Amount: INR 2,700,000 Payment type: SETTLEMENT Payment status: SUCCESS

Payment date: 2026-10-09 Created at: 2026-10-09 10:23:20

Narrative: The payment table records a successful settlement payment of

INR 2.7 million. The amount matches settlement 1 and the recommended settlement for claim 1.

DATABASE RECORD: PAYMENT 3 — CLAIM 2 SETTLEMENT Payment ID: 3 Policy ID:

1 Amount: INR 66,666.24 Payment type: SETTLEMENT Payment status: SUCCESS

Payment date: 2026-10-09 Created at: 2026-10-09 12:25:34

Narrative: The payment table records a successful settlement payment of

INR 66,666.24. The amount matches settlement 2 and the recommended settlement for claim 2. Because claim 2 appears to contain placeholder data, verify the record before treating it as a real-world insurance payment.

Payment concepts: - PREMIUM: A payment associated with purchasing or

maintaining the insurance policy. The supplied record is INR 10 million

and SUCCESS. - SETTLEMENT: A payment associated with a claim settlement.

The supplied records include INR 2.7 million and INR 66,666.24, both marked SUCCESS. - SUCCESS is the status recorded in the payments table. It is not a statement about any external banking system beyond the supplied record. - A payment record, settlement record, claim record, and policy record are related but distinct records. They should be cross-checked by their IDs and amounts when answering payment-status questions.

16. NOTIFICATIONS AND SYSTEM EVENTS

The supplied notifications table contains 28 records. Each record has a notification ID, recipient user ID, title, message, read flag, and timestamp. In this database, is_read = 1 means read, and is_read = 0 means unread.

The descriptions below retain all 28 records and map recipient IDs to the users table. The timestamps are database creation timestamps, not necessarily the dates described in an inspection's inspection_date field.

1. PROJECT CREATED Notification ID: 1 Recipient: User ID 2 — Road

Authority PM Title: Project Created Message: "Road project NH 47

Road Expansion has been created successfully." Read status: READ

Created at: 2026-10-09 09:53:31

Meaning: The road authority user received a notification confirming that

the NH 47 Road Expansion project was created.

2. CONSTRUCTION DETAILS SUBMITTED Notification ID: 2 Recipient: User ID

4 — Contractor PM Title: Construction Details Submitted Message:

"Construction details submitted for project NH 47 Road Expansion."

Read status: UNREAD Created at: 2026-10-09 09:55:22

Meaning: The notification confirms that construction details were

submitted for NH 47. The recorded recipient is the Contractor PM. The message and recipient are preserved as they appear in the database.

3. INSPECTION CREATED Notification ID: 3 Recipient: User ID 7 — Road

Inspector Title: Inspection Created Message: "Inspection has been

created successfully." Read status: UNREAD Created at: 2026-10-09

09:56:06

Meaning: The inspector received confirmation that an inspection record

was created.

4. INITIAL INSPECTION APPROVED Notification ID: 4 Recipient: User ID 2

— Road Authority PM Title: Initial Inspection Approved Message:

"Initial inspection has been approved successfully." Read status:

READ Created at: 2026-10-09 09:56:13

Meaning: The road authority was notified that the initial inspection had

been approved.

5. CONSTRUCTION PROGRESS UPDATED TO 20% Notification ID: 5 Recipient:

User ID 4 — Contractor PM Title: Construction Progress Updated

Message: "Construction progress updated to 20% for project ID 1."

Read status: UNREAD Created at: 2026-10-09 09:57:29

Meaning: A notification records the 20% progress update for project 1.

6. CONSTRUCTION PROGRESS UPDATED TO 40% Notification ID: 6 Recipient:

User ID 4 — Contractor PM Title: Construction Progress Updated

Message: "Construction progress updated to 40% for project ID 1."

Read status: UNREAD Created at: 2026-10-09 09:57:53

Meaning: A notification records the 40% progress update for project 1.

7. INSPECTION CREATED Notification ID: 7 Recipient: User ID 7 — Road

Inspector Title: Inspection Created Message: "Inspection has been

created successfully." Read status: UNREAD Created at: 2026-10-09

09:58:51

Meaning: The inspector received confirmation that another inspection record was created.

8. DEFECT REPORTED Notification ID: 8 Recipient: User ID 7 — Road **Inspector Title:** Defect Reported Message: “A new defect has been **reported successfully.” Read status:** UNREAD Created at: 2026-10-09 **09:**59:26

Meaning: A defect-reporting event generated a notification to the Road Inspector. The defect record identifies a material-quality problem linked to inspection 2.

9. CORRECTIVE ACTION SUBMITTED Notification ID: 9 Recipient: User ID 4 **— Contractor PM Title:** Corrective Action Submitted Message: **“Corrective action has been submitted successfully.” Read status: UNREAD Created at:** 2026-10-09 10:03:27

Meaning: A notification was created for the contractor-side user when corrective action was submitted. The corrective-action record states that the material was changed.

10. INSPECTION REJECTED Notification ID: 10 Recipient: User ID 2 — Road **Authority PM Title:** Inspection Rejected Message: “Inspection has **been rejected and rework is required.” Read status:** READ Created at: **2026-10-09 10:**03:48

Meaning: The Road Authority PM was notified that an inspection was rejected and rework was required. This corresponds to the failed periodic inspection scenario in the NH 47 records.

11. CORRECTIVE ACTION VERIFIED Notification ID: 11 Recipient: User ID 7 **— Road Inspector Title:** Corrective Action Verified Message: **“Corrective action verification has been completed.” Read status: UNREAD Created at:** 2026-10-09 10:04:02

Meaning: The inspector received confirmation that corrective-action verification was completed. The database’s corrective-action record has verification status VERIFIED.

12. INSPECTION CREATED Notification ID: 12 Recipient: User ID 7 — Road **Inspector Title:** Inspection Created Message: “Inspection has been **created successfully.” Read status:** UNREAD Created at: 2026-10-09 **10:**06:02

Meaning: A notification confirms creation of another inspection record.

13. PERIODIC INSPECTION APPROVED Notification ID: 13 Recipient: User ID **2 — Road Authority PM Title:** Periodic Inspection Approved Message: **“Periodic inspection has been approved successfully.” Read status: READ Created at:** 2026-10-09 10:08:27

Meaning: The Road Authority PM was notified of an approved periodic inspection.

14. CONSTRUCTION PROGRESS UPDATED TO 75% Notification ID: 14 Recipient: **User ID 4 — Contractor PM Title:** Construction Progress Updated **Message:** “Construction progress updated to 75% for project ID 1.”

Read status: UNREAD Created at: 2026-10-09 10:09:33

Meaning: A notification records the 75% construction progress update.

15. INSPECTION CREATED Notification ID: 15 Recipient: User ID 7 — Road **Inspector Title:** Inspection Created Message: “Inspection has been **created successfully.” Read status:** UNREAD Created at: 2026-10-09 **10:**11:19

Meaning: The inspector received confirmation that an inspection record was created.

16. PERIODIC INSPECTION APPROVED Notification ID: 16 Recipient: User ID **2 — Road Authority PM Title:** Periodic Inspection Approved Message: **“Periodic inspection has been approved successfully.” Read status: READ Created at:** 2026-10-09 10:11:39

Meaning: The Road Authority PM received a second notification for an approved periodic inspection.

17. CONSTRUCTION PROGRESS UPDATED TO 100% Notification ID: 17 Recipient: **User ID 4 — Contractor PM Title:** Construction Progress Updated **Message:** “Construction progress updated to 100% for project ID 1.” **Read status:** UNREAD Created at: 2026-10-09 10:12:04

Meaning: The notification records the final 100% construction progress update for project 1. Progress completion should still be distinguished from any separate contractual or insurance status.

18. INSPECTION CREATED Notification ID: 18 Recipient: User ID 7 — Road **Inspector Title:** Inspection Created Message: “Inspection has been **created successfully.” Read status:** UNREAD Created at: 2026-10-09 **10:**12:44

Meaning: The inspector received confirmation that the final inspection record was created.

19. FINAL INSPECTION APPROVED Notification ID: 19 Recipient: User ID 2 — **Road Authority PM Title:** Final Inspection Approved Message: “Final **inspection has been approved successfully.” Read status:** READ **Created at:** 2026-10-09 10:17:22

Meaning: The Road Authority PM received a notification confirming approval of the final inspection.

20. RISK ASSESSMENT CREATED Notification ID: 20 Recipient: User ID 5 — **Insurance Underwriter Title:** Risk Assessment Created Message: “Risk assessment has been created successfully for project ID 1.” Read **status:** UNREAD Created at: 2026-10-09 10:18:33

Meaning: The Insurance Underwriter received confirmation that a risk assessment was created for NH 47.

21. POLICY PROPOSAL GENERATED Notification ID: 21 Recipient: User ID 5 — **Insurance Underwriter Title:** Policy Proposal Generated Message:

“Insurance policy proposal has been generated successfully.” Read

status: UNREAD Created at: 2026-10-09 10:18:56

Meaning: The underwriter received a notification that a policy proposal

had been generated. The supplied proposal record is ACCEPTED.

22. POLICY ACTIVATED Notification ID: 22 Recipient: User ID 2 — Road

Authority PM Title: Policy Activated Message: “Policy POL-11E7B1AB

has been activated and premium payment has been processed

successfully.” Read status: READ Created at: 2026-10-09 10:19:12

Meaning: The Road Authority PM was notified that policy POL-11E7B1AB was

activated and the premium payment was processed successfully. The policy is recorded as ACTIVE, and the premium payment is marked SUCCESS.

23. CLAIM SUBMITTED Notification ID: 23 Recipient: User ID 2 — Road

Authority PM Title: Claim Submitted Message: “Your insurance claim

has been submitted successfully.” Read status: READ Created at:

2026-10-09 10:21:25

Meaning: The Road Authority PM received a claim-submission notification.

The recipient and event are reproduced from the database, even though the recipient is also the recorded user who raised the claim.

24. CLAIM ASSESSED Notification ID: 24 Recipient: User ID 6 — Claims

Adjuster Title: Claim Assessed Message: “Claim assessment has been

completed successfully.” Read status: READ Created at: 2026-10-09

10:21:51

Meaning: The Claims Adjuster received confirmation that a claim

assessment was completed. Claim assessment 1 is recorded as COMPLETED.

25. CLAIM APPROVED Notification ID: 25 Recipient: User ID 6 — Claims

Adjuster Title: Claim Approved Message: “Claim approved and

settlement payment processed successfully.” Read status: READ

Created at: 2026-10-09 10:23:20

Meaning: The Claims Adjuster received a notification stating that the

claim was approved and the settlement payment was processed. The corresponding settlement is INR 2.7 million and the payment status is PAID/SUCCESS in the settlement and payment records.

26. CLAIM SUBMITTED Notification ID: 26 Recipient: User ID 2 — Road

Authority PM Title: Claim Submitted Message: “Your insurance claim

has been submitted successfully.” Read status: READ Created at:

2026-10-09 12:21:54

Meaning: A second claim-submission notification was recorded for the

Road Authority PM. It corresponds in time to claim 2, whose reason appears to be placeholder data.

27. CLAIM ASSESSED Notification ID: 27 Recipient: User ID 6 — Claims

Adjuster Title: Claim Assessed Message: “Claim assessment has been

completed successfully.” Read status: UNREAD Created at: 2026-10-09

12:25:22

Meaning: A claim-assessment notification was created for the Claims

Adjuster. It corresponds to assessment 2, which is marked COMPLETED but contains the placeholder-looking note “fsgdg.”

28. CLAIM APPROVED Notification ID: 28 Recipient: User ID 6 — Claims

Adjuster Title: Claim Approved Message: “Claim approved and

settlement payment processed successfully.” Read status: UNREAD

Created at: 2026-10-09 12:25:34

Meaning: The final supplied notification records claim approval and

settlement processing for the second claim. The matching settlement and payment records show INR 66,666.24. The claim’s placeholder-looking source data should be verified before this record is used as a real insurance example.

NOTIFICATION SUMMARY The 28 records cover project creation, construction detail submission, inspection creation and approval, construction progress updates, defect reporting, corrective-action submission and verification, risk assessment, policy proposal generation, policy activation, claim submission, claim assessment, and claim approval.

The database contains both read and unread notifications. A read flag reports whether the notification has been marked read; it does not change the underlying project, inspection, policy, claim, or payment status.

The notification table’s recorded recipients should be preserved when describing actual events. Some recipients or event titles may appear unusual from a workflow-design perspective. Do not silently rewrite those database records. A production implementation may choose different recipient rules, but that should be documented as a proposed change rather than as an existing fact.

17. END-TO-END BUSINESS WORKFLOWS

WORKFLOW A — ORGANIZATION AND USER SETUP 1. An administrator creates or maintains organization records. 2. User accounts are associated with organizations and assigned roles. 3. Role-based access determines which actions a user may perform. 4. Organization association helps restrict project access to the relevant road authority, contractor, insurer, or inspection organization. 5. Password hashes and authentication secrets remain in the authentication system and are not exposed in the RAG knowledge base.

WORKFLOW B — ROAD PROJECT CREATION 1. The Road Authority PM creates a road project. 2. The project is associated with a road authority organization and a contractor organization. 3. Project details include name, description, location, road length, estimated cost, expected lifespan, start date, and estimated completion date. 4. A project-created notification may be generated. 5. The contractor submits construction details such as material type, methodology, timeline, equipment, and workforce. 6. The project status and construction progress are tracked separately.

WORKFLOW C — CONSTRUCTION PROGRESS 1. The Contractor PM records progress percentage and a work description. 2. Each update should be associated with the relevant project and the user who created it. 3. A progress notification may be generated. 4. The Road Authority PM reviews progress and investigates material delays, unexplained jumps, or conflicting status information. 5. A progress value of 100% indicates reported completion but does not by itself establish that every inspection or contractual approval is complete.

WORKFLOW D — INSPECTION AND DEFECT MANAGEMENT 1. The Road Inspector creates an initial, periodic, or final inspection. 2. The inspector records material-quality, layer-thickness, and test scores, remarks, inspection status, and approval status. 3. If the inspection fails, the inspector documents the finding and creates or links a defect record. 4. A notification may inform relevant participants that an inspection was rejected and rework is required. 5. The contractor investigates the issue and records a corrective action. 6. The inspector verifies the corrective action and records the outcome. 7. A follow-up inspection may confirm that the issue is resolved. 8. The defect should only be described as verified or resolved when the corresponding records support that conclusion.

WORKFLOW E — RISK ASSESSMENT AND POLICY PROPOSAL 1. The Insurance Underwriter reviews project details, construction records, inspection results, defects, and other available risk factors. 2. A risk assessment stores the component scores, risk score, category, recommended coverage, and recommended premium. 3. The underwriting rules should be applied consistently. The exact scoring formula must come from the application configuration; it is not supplied in the database output. 4. A policy proposal is generated and records premium, coverage, validity, status, and creator. 5. The proposal may be accepted or rejected according to the configured business process. 6. A proposal should not be described as an active policy until an issued policy record confirms the policy status.

WORKFLOW F — POLICY ACTIVATION AND PREMIUM PAYMENT 1. An accepted proposal is used to create an issued policy according to the application’s workflow. 2. The policy record stores the policy number, project, premium, coverage,

issue date, expiry date, and status. 3. The premium payment is recorded with its type, amount, date, and status. 4. A policy-activation notification may be generated after the required activation and payment conditions are satisfied. 5. Policy status, premium payment status, and coverage terms should be checked separately when answering questions.

WORKFLOW G — CLAIM SUBMISSION 1. The Road Authority or government-side policyholder reports a potential insured loss under a relevant policy.

2. The claim identifies the project, policy, reason, estimated repair

cost, raising user, status, and any rejection reason. 3. The claim should include evidence such as photographs, inspection reports, repair estimates, location, dates, and relevant material or construction records where available. 4. Claim submission does not mean that coverage has been accepted. 5. A notification may confirm receipt of the claim.

WORKFLOW H — CLAIM ASSESSMENT 1. The Claims Adjuster reviews the claim and verifies the policy period. 2. The adjuster examines the cause of damage and compares it with covered causes and exclusions. 3. The adjuster evaluates evidence, severity, and eligible loss. 4. The assessment records notes, severity, recommended settlement, status, and adjuster identity. 5. The adjuster should not infer that heavy traffic, weather, or a defect automatically makes the loss covered. The cause must be established and compared with the actual policy terms. 6. A claim assessment notification may be generated.

WORKFLOW I — SETTLEMENT AND PAYMENT 1. Following the configured approval process, a settlement record is created. 2. The settlement records the claim, settlement amount, payment status, approver, and timestamp. 3. A payment record may separately record the settlement transaction. 4. The settlement amount may differ from the estimated repair cost because the adjuster may determine that only part of the claimed cost is eligible.

5. The assistant should use the recorded assessment and policy wording

when explaining the difference. If no calculation is recorded, it should not invent one. 6. Claim status, settlement status, and payment status should be cross-checked. 7. A claim should not be called paid merely because a claim-assessment record is complete.

WORKFLOW J — NOTIFICATION HANDLING 1. When a supported system event occurs, the application creates a notification with a title, message, recipient, read flag, and timestamp. 2. Notifications can inform users about project creation, construction submissions, progress updates, inspections, defects, corrective actions, risk assessments, policy proposals, policy activation, claims, assessments, and settlements. 3. The read flag indicates whether a notification has been read. It is not the same as the business record’s status. 4. Notification recipients should follow the configured authorization and workflow rules. Existing database recipients should be reported exactly when answering questions about the supplied records.

18. IMPORTANT BUSINESS DEFINITIONS

Road Authority: The government-side organization responsible for the

road project. In this RoadSure model, the road authority is the policyholder-side participant that may raise a claim for a covered road failure.

Contractor: The organization responsible for executing construction or

rehabilitation work, submitting construction details, reporting progress, and responding to inspection findings.

Inspection: A technical review of road work or materials. Inspections

may be initial, periodic, or final. Their recorded scores and outcomes help document quality.

Defect: A documented problem found during construction or inspection. A

defect may require corrective action but is not automatically an insured loss.

Corrective action: Work performed to address a defect. The action and

its verification status should be recorded.

Risk assessment: An underwriting record containing risk-related scores,

a risk category, recommended coverage, and recommended premium.

Policy proposal: A proposed insurance arrangement. It is distinct from

an issued policy.

Policy: The issued insurance record containing identifiers, premium,

coverage, status, and validity dates.

Claim: A request for insurance benefits for a reported loss. Submission

does not guarantee acceptance or payment.

Claim assessment: The adjuster's evaluation of the claim, including

severity, notes, and recommended settlement.

Settlement: The recorded amount approved or otherwise processed for a

claim according to the application's workflow.

Payment: A transaction record identifying amount, type, status, and

date. Premium payments and claim settlement payments are different payment types.

Notification: A message generated for a user in response to a system

event. A notification's read status is independent of the business event's actual status.

Premature road failure: Road deterioration or failure occurring earlier

than reasonably expected and attributable to a cause that may be covered by the relevant policy. Whether a particular failure is covered depends on evidence and policy terms.

Normal wear and tear: Expected deterioration arising from ordinary use

and time. It should not automatically be treated as an insured construction defect.

Routine maintenance: Ordinary upkeep and recurring maintenance work. It

should not automatically be treated as an insured loss.

19. DATA QUALITY AND RETRIEVAL RULES

1. Preserve actual records. When asked about the supplied database,

report the stored value. Do not silently replace it with a more plausible value.

2. Identify sample records. Projects 2 and 3 and their associated

example progress, inspection, risk, and policy scenarios are illustrative additions in this file. They must not be described as verified live database rows.

3. Flag inconsistent chronology. NH 47 is marked COMPLETED and has 100%

progress, yet its estimated completion date is 2029-07-25 and its inspection dates extend into 2027. Report these values as recorded and flag the discrepancy when relevant.

4. Flag placeholder claim data. Claim 2 has the reason "12121212112";

assessment 2 has the note "fsgdg". These records look like test data. Do not invent the underlying damage cause.

5. Do not expose secrets. Passwords, password hashes, tokens, or

authentication secrets are not knowledge-base content.

6. Do not invent scoring formulas. The risk score is stored as 18.67

and category LOW. The formula is not available in the supplied data.

7. Do not overstate insurance coverage. A coverage percentage does not

mean that every repair bill is payable at that percentage. Coverage depends on policy terms and claim assessment.

8. Distinguish statuses. Project status, progress percentage,

inspection status, inspection approval, defect status, corrective-action verification, proposal status, policy status, claim status, assessment status, settlement status, payment status, and notification read status are different concepts.

9. Cross-check related records. For payment questions, compare claims,

assessments, settlements, and payments. For inspection questions, compare inspection, defect, corrective action, and follow-up inspection records.

10. Avoid unsupported conclusions. If the knowledge base does not

contain an answer, say that the available records do not specify it. Do not fabricate official rules or evidence.

20. FREQUENTLY ASKED QUESTIONS — 30 QUESTIONS AND ANSWERS

FAQ 1. What is RoadSure? RoadSure is a road-quality insurance platform

that connects road authorities, contractors, inspection organizations, and an insurer. It records road projects, construction details, progress, inspections, defects, corrective actions, risk assessments, policies, claims, settlements, payments, and notifications.

FAQ 2. What kinds of road failures is RoadSure intended to cover?

RoadSure is intended to cover eligible premature road failures attributable to covered construction, material, design, or quality risks. The exact covered causes and exclusions depend on the policy. Normal wear and tear and routine maintenance should not be assumed to be covered.

FAQ 3. Does RoadSure cover every pothole or road repair? No. A pothole

or repair need alone does not establish insurance coverage. The cause, timing, evidence, policy wording, exclusions, and claim assessment must be considered. A defect may be a maintenance issue, normal wear, or an uncovered event.

FAQ 4. Which organizations are in the supplied database? The database

contains Government Road Authority, ABC Infrastructure Pvt Ltd, Hartford Insurance, Road Quality Labs, and RoadCorp Contractors. All five organization records are marked ACTIVE.

FAQ 5. Who is the policyholder-side participant in the RoadSure

workflow? Government Road Authority is the road authority in the supplied data. The Road Authority PM raises claims in the recorded NH 47 examples. A contractor should not be assumed to be the policyholder or settlement recipient.

FAQ 6. What are the main user roles? The supplied roles are ADMIN,

ROAD_AUTHORITY_PM, CONTRACTOR_PM, UNDERWRITER, CLAIMS_ADJUSTER, and INSPECTOR. Each role has a different responsibility covering platform administration, project management, construction, risk assessment, claim adjustment, or technical inspection.

FAQ 7. Who created the NH 47 Road Expansion project? The NH 47 project

was created by user ID 2, the Road Authority PM, on 2026-10-09 at

09:53:29.

FAQ 8. What is the NH 47 Road Expansion project? It is a 100-kilometre

project in Chennai intended to expand a capacity-limited highway to eight lanes. The recorded estimated cost is INR 1 billion, and the expected lifespan is 50 years. ABC Infrastructure Pvt Ltd is the contractor and Government Road Authority is the road authority.

FAQ 9. What is the recorded status of NH 47? The database records its

project status as COMPLETED. Construction progress records reach 100%. The database also records an estimated completion date of 2029-07-25, so the discrepancy between status and date should be flagged rather than silently resolved.

FAQ 10. What construction material is recorded for NH 47? The

construction details record lists ASPHALT as the material type. It also describes subgrade stabilization, aggregate base installation, asphalt paving, compaction, and final quality inspections.

FAQ 11. What are the NH 47 progress milestones? The supplied progress

history records 20% when excavation was completed, 40% when the pipeline had been laid out, 75% when road laying started, and 100% when the road was reported completed. All four records are dated 2026-10-09.

FAQ 12. How many inspections are recorded for NH 47? Five inspections

are recorded: one initial inspection, three periodic inspections, and

one final inspection.

FAQ 13. Did the initial NH 47 inspection pass? Yes. Inspection 1 is

recorded as PASSED and APPROVED. Its material-quality score is 90, layer-thickness score is 85, and test score is 92.

FAQ 14. What happened during the failed periodic inspection? Inspection

2 failed and was rejected. It recorded material quality of 40, layer thickness of 90, and test score of 50. The remarks say that the new material did not meet test standards. A material defect was linked to this inspection.

FAQ 15. What defect was recorded for NH 47? Defect 1 is a Material

Defect described as “Material quality is bad.” It is linked to project 1 and inspection 2. The defect status is RESOLVED.

FAQ 16. How was the NH 47 material defect addressed? The

corrective-action record says the material was changed. The verification status is VERIFIED. A subsequent periodic inspection is also recorded as passed and approved, with remarks stating that the defect had been resolved.

FAQ 17. Did the final NH 47 inspection pass? Yes. Inspection 5 is

recorded as PASSED and APPROVED. Its material-quality score is 90, layer-thickness score is 90, and test score is 92. The remarks say that all tests passed with standard results.

FAQ 18. What is the NH 47 risk category? The stored risk assessment

classifies NH 47 as LOW risk, with a risk score of 18.67. The database does not provide the formula used to calculate that score.

FAQ 19. What coverage and premium were recommended for NH 47? The risk

assessment recommends 90% coverage and a premium of INR 10 million. The policy proposal and issued policy records also store 90% coverage and INR 10 million as the premium.

FAQ 20. What is the policy number for NH 47? The policy number is

POL-11E7B1AB. The policy record is ACTIVE, with issue date 2026-10-09 and expiry date 2027-10-09.

FAQ 21. Does 90% coverage guarantee payment of 90% of every repair bill?

No. The database stores 90% as the coverage percentage, but that alone does not establish how every claim is calculated. Actual payment depends on the policy wording, covered cause, exclusions, eligible loss, limits or deductibles if applicable, and claim assessment.

FAQ 22. Who can raise a claim in the supplied NH 47 records? User ID 2,

the Road Authority PM associated with Government Road Authority, raised both recorded claims. This reflects the supplied records and the policyholder-side workflow used in this knowledge base.

FAQ 23. What is claim 1 about? Claim 1 reports potholes attributed in

its reason to heavy traffic. Its estimated repair cost is INR 10 million, and its status is SETTLED. Its assessment classifies the severity as MINOR and recommends INR 2.7 million.

FAQ 24. How much was settled for claim 1? The settlement record shows

INR 2.7 million with payment status PAID. The payment table contains a matching INR 2.7 million settlement payment marked SUCCESS on 2026-10-09.

FAQ 25. Why is the claim 1 settlement lower than its estimated repair

cost? The database records an estimated repair cost of INR 10 million and a recommended settlement of INR 2.7 million, but the supplied assessment notes do not provide a detailed calculation. The assistant should not invent the reason. In a real claim, the difference would need to be explained using the adjuster's evidence, eligible loss, and policy terms.

FAQ 26. Is claim 2 reliable as a real-world example? Claim 2 is a

database record marked SETTLED, but its reason is "12121212112" and its assessment note is "fsgdg." Those values appear to be placeholders or test data. The record should be preserved for traceability but verified before being used as a genuine insurance example.

FAQ 27. What payments are recorded? Three payments are recorded: a

successful premium payment of INR 10 million, a successful claim settlement payment of INR 2.7 million, and a successful settlement payment of INR 66,666.24. All are dated 2026-10-09.

FAQ 28. What types of notifications does RoadSure generate in the

supplied data? The 28 records include notifications for project creation, construction details submission, inspection creation and approval, progress updates, defect reporting, corrective-action submission and verification, risk assessment, policy proposal generation, policy activation, claim submission, claim assessment, and claim approval.

FAQ 29. What does a notification marked unread mean? In the supplied

notifications table, is_read = 0 means UNREAD and is_read = 1 means READ. This field describes whether the notification was read. It does not determine whether a project, inspection, policy, claim, or payment is complete or successful.

FAQ 30. What should the assistant do when information is missing or

inconsistent? It should report the stored values, identify the inconsistency, and explain what the available records do and do not establish. It should distinguish database facts from illustrative sample data, avoid exposing secrets, avoid inventing risk formulas or policy rules, and state when a detail requires verification in the source system.

21. ADDITIONAL SAMPLE QUESTIONS FOR FUTURE RAG TESTING

The following prompts can be used to test whether a RAG system retrieves the right sections and respects distinctions between stored facts and

illustrative examples:

- Which organization employs the Road Inspector?
- Which user is associated with the Claims Adjuster role?

- What equipment and workforce are listed for NH 47?
- Which inspection is linked to defect 1?
- What was the status of the corrective action?
- Which user was recorded as generating policy proposal 1?
- What is the policy expiry date?
- What is the status of the premium payment?
- Which notification confirms policy activation?
- Which notifications are unread?
- What was the settlement amount for claim 2?
- What is the recorded note for claim assessment 2?
- Which fields should be checked before deciding whether pothole damage is covered?
- Why should a 100% progress value not automatically be treated as policy activation?
- What additional evidence would help an adjuster assess premature pavement failure?
- What is the difference between a policy proposal and an issued policy?
- What is the difference between a claim assessment and a settlement?
- How should the system handle a failed inspection?
- Why should a material defect not automatically be treated as an insured loss?
- Which facts about projects 2 and 3 are illustrative rather than verified live database records?

22. FINAL REFERENCE SUMMARY

Confirmed database facts: - 5 organizations and 7 users are supplied. -

NH 47 Road Expansion is project ID 1 and is recorded as COMPLETED. - NH 47 has one construction-detail record and four progress records. - Five inspections are recorded for NH 47. - One material defect and one verified corrective action are recorded. - One risk assessment is recorded with score 18.67 and category LOW. - One accepted policy proposal is recorded. - One active policy is recorded as POL-11E7B1AB. - Two claims, two claim assessments, two settlements, and three payment records are supplied. - The notifications table contains 28 records. - Password hashes are intentionally excluded from this document.

Illustrative additions: - Project 2: Outer Ring Road Smart Corridor in

Hyderabad. - Project 3: Coastal Highway Rehabilitation in

Visakhapatnam. - Example construction details, progress scenarios, risk scores, coverage values, premiums, and inspection scenarios for projects 2 and 3. - Proposed workflows and recommendations that go beyond the explicit database fields.

Key caution: RoadSure should be described as a cause-based road-quality

insurance platform. A claim should be evaluated against the relevant policy and evidence. Do not state that every road defect, pothole, weather event, traffic-related issue, or repair is automatically covered.